

ESWIN

EIEBX10-A30(IWB) User Manual

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2026/05/06

Notice

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Change History

Version	Date	Description
v1.0	2026/05/06	Added local management system
v0.5	2025/10/28	Initial version

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1. Overview

1.1. Product Introduction

The EIEBXIO-A30 (IWB) Intelligent Analysis Station is powered by the EIC7700 RISC-V chip, independently developed by ESWIN, delivering computing power of up to 13.3 TOPS (INT8).

Key Features:

- **Audio/Video Input:** Supports connection to external pick-up devices (Phoenix terminals) and network IPC cameras for audio and video stream capture.
- **AI Behavior Analysis:** Analyzes captured audio and video, including behavior, face detection, and keypoint detection.
- **Cloud Synergy:** Transmits analysis results to the cloud via HTTP protocol, allowing users to view reports by logging into their cloud account via the terminal.
- **Flexible Power Supply:** Supports both POE and DC adapter power supply to adapt to various environments.

1.2. Technical Specifications

Table 1: Technical Specifications

Category	Specifications
Mechanical Characteristics	Dimensions: 157 x 110 x 30 mm
	Enclosure Material: Metal chassis (1mm galvanized sheet)
	Color: Black
	Mounting Method: Surface Mount / Flush Mount
	SN Code: Unique factory SN code
Performance Parameters	Processor: ESWIN EIC7700 RISC-V RV64GC 4 cores @1.4Ghz up to 1.8Ghz
	Memory & Storage: 16 GB LPDDR5 + 32 GB eMMC Flash
	AI Computing Power: 13.3 TOPS (INT8)
	Power Consumption: Typical 20 W (POE bt standard, Adapter MAX 12V5A)
	Operating System: Debian System
	Cooling Method: Active Cooling (Heatsink specs: 57×83×15.7 mm, AL6063-T5 black anodized)
	Video Codec: Supports HEVC (H.265) and AVC (H.264) encoding/decoding H.265 Decoding: up to 8K@50fps or 32-channel 1080p@25fps H.265 Encoding: up to 8K@25fps or 16-channel 1080p@25fps
	Audio Codec Type: AAC-LC encoding G.711/G.722.1/G.726/MP2L2/PCM/MP3/AAC-LC decoding
Interface Parameters	Network Interface: 2 x RJ45 10/100/1000Mbps adaptive
	USB Interface: 2 x USB 3.0
	Debug Port: 1 x USB Type-C
	Phoenix Terminal: Supports connection of two pick-up devices
	Storage Card Interface: 1 x TF card 1 x M.2 2280 PCIeGen3.0 nvme
	Power Interface: 1 x 5521 DC IN @ 12V5A, PD IN

Table 1: Technical Specifications

Category	Specifications
Environmental Characteristics	Operating Temperature: -20 °C to 60 °C
	Storage Temperature: -40 °C to 85 °C
	Relative Operating Humidity: 5%RH - 90%RH

1.3. Appearance & Structure



Figure 1: Product Image

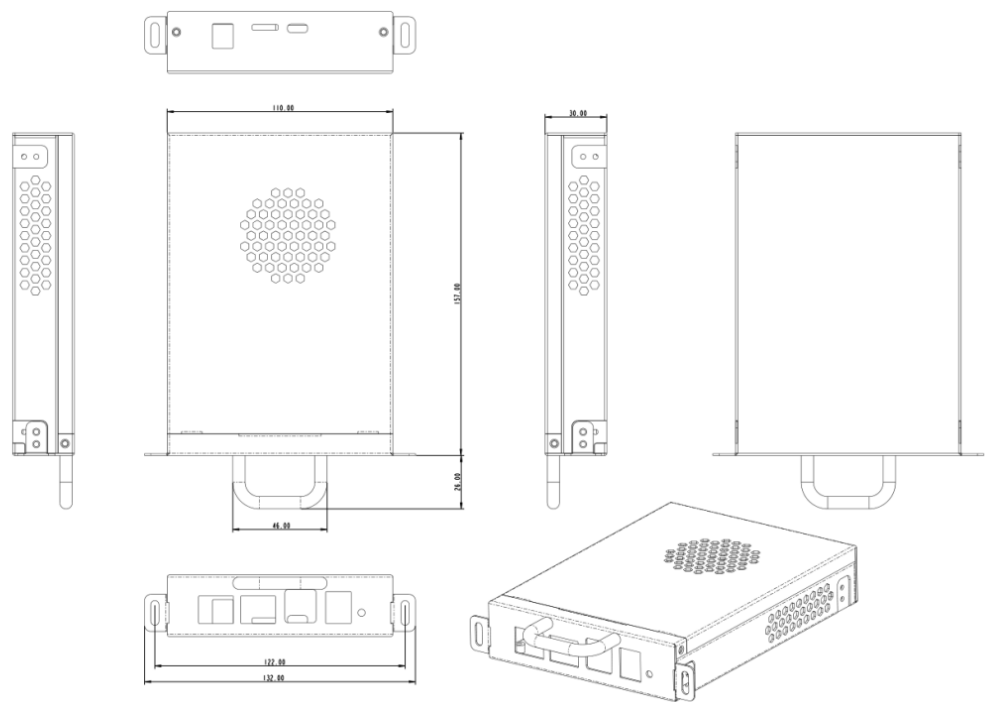


Figure 2: Product Six-View Diagram

1.4. OS Versions

OS Name	Kernel Version	Support Status	Maintenance Status
Debian	6.6	Supported	Primary Maintenance

1.5. Development Resources

To facilitate secondary development, algorithm porting, and system integration, ESWIN provides comprehensive documentation and developer community support.

Resource	Description
Developer Forum	A technical exchange forum where users can ask questions, share experiences, and get the latest updates.
Resource Center	Centralized access to flashing images, ESSDK, BSP, domain-specific development manuals, and quick start guides.
ESSDK Quick Start Guide	Core development documentation for the ESSDK platform, covering model quantization, compilation, and on-board deployment processes.

2. Interfaces

2.1. Front Interfaces

Interface Name	Description
Network Interface	2 x RJ45 10/100/1000Mbps adaptive (supports port switching)
Power Interface (PD IN)	Supports POE power supply and PD protocol input
USB Interface	2 x USB 3.0
Phoenix Terminal	Supports connection of two pick-up devices (microphones)
Power Indicator	Solid on when powered

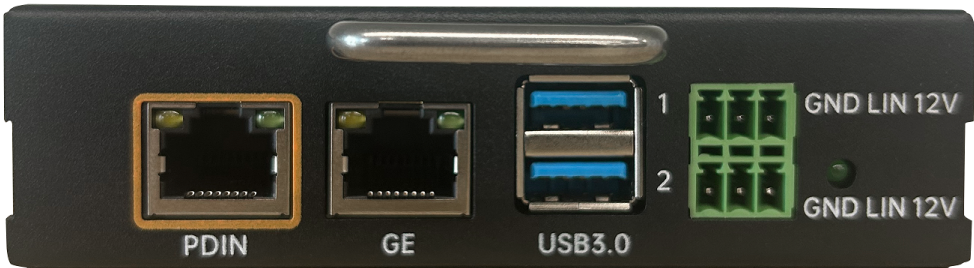


Figure 3: Front Interfaces

Tip

After system boot: Both USB1 and USB2 function as standard interfaces for connecting peripherals such as USB flash drives, keyboards, and mice **In Bootloader mode:**

- USB1 (Device): Dedicated to fastboot flashing; cannot be used for connecting peripherals.
- USB2 (Host):Used for connecting peripherals; does not support fastboot flashing.

2.2. Rear Interfaces

Interface Name	Description
USB Interface (Debug Port)	1 x USB Type-C (Serial Debug)
TF Card Slot	TF card slot for storage expansion
DC Power Interface	1 x 5521 DC IN @ 12V5A

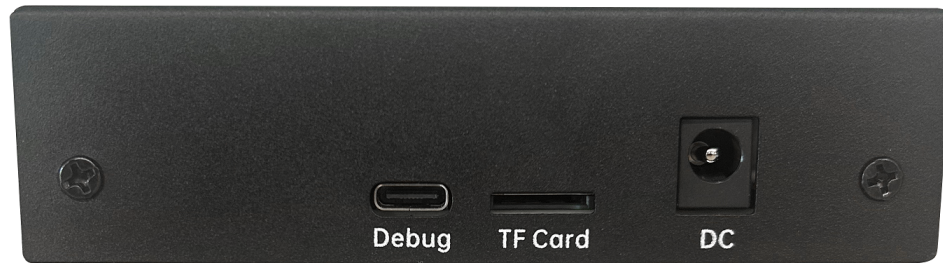


Figure 4: Rear Interfaces

3. User Guide

This section guides users through the power-on process, debugging connections, and basic operations of the Local Management System for the EIEBXIO-A30 (IWB) device.

3.1. Power On

The IWB device supports two power supply methods. Please choose one based on your on-site environment:

1. **PoE Power Supply**
 - Connect the device's **orange PDIN port** to a PoE switch using an Ethernet cable.
 - **Status Check:** A solid power indicator light means the device is powered on successfully; a flashing orange light on the port indicates an active network connection.
2. **Power Adapter Supply**
 - Connect the standard 12V/5A power adapter to the DC power interface on the back of the device.
 - **Status Check:** A solid power indicator light means the device is powered on successfully.
 - Connect the device's GE port to a standard switch or router using an Ethernet cable.

3.2. Device Debugging

The IWB provides both SSH and serial port debugging methods for developers or operations personnel to perform in-depth maintenance.

3.2.1. SSH Remote Login

1. **Network Configuration:** Ensure the PC and the IWB are on the same local area network (default subnet: 192.168.1.x).
2. **Login Credentials:**
 - **IP Address:** Default is 192.168.1.36
 - **Username:** eswin
 - **Password:** eswin
3. **Connection Command:**

```
ssh eswin@192.168.1.36
```

3.2.2. Serial Debugging (Type-C)

When the network is unreachable or the system is abnormal, the serial port can be used for physical debugging.

1. **Driver Installation:** On a Windows PC, you must first install the CH340/CH341 USB-to-serial driver, link: [CH340/CH341 driver](#).
2. **Hardware Connection:** Use a Type-C data cable to connect the PC to the debug port on the front of the IWB.
3. **Software Configuration:** Use a terminal tool (such as MobaXterm or PuTTY), select the identified COM port, and configure the parameters as **115200 8N1**.

3.3. Local Management System (Web UI)

In addition to the command line, the IWB provides a Web-based graphical management interface for quick on-site configuration and status monitoring.

3.3.1. Network Preparation and Login

1. **Network Preparation:** Ensure the PC and the IWB are on the same local area network.
2. **PC IP Configuration:** If manual configuration is required, set the PC' s IP to 192.168.1.x (e.g., 192.168.1.100) and the subnet mask to 255.255.255.0 .
3. **System Login:**
 - Open a browser and enter http://192.168.1.36 .
 - Enter the default credentials: Username admin , Password Admin@123 .

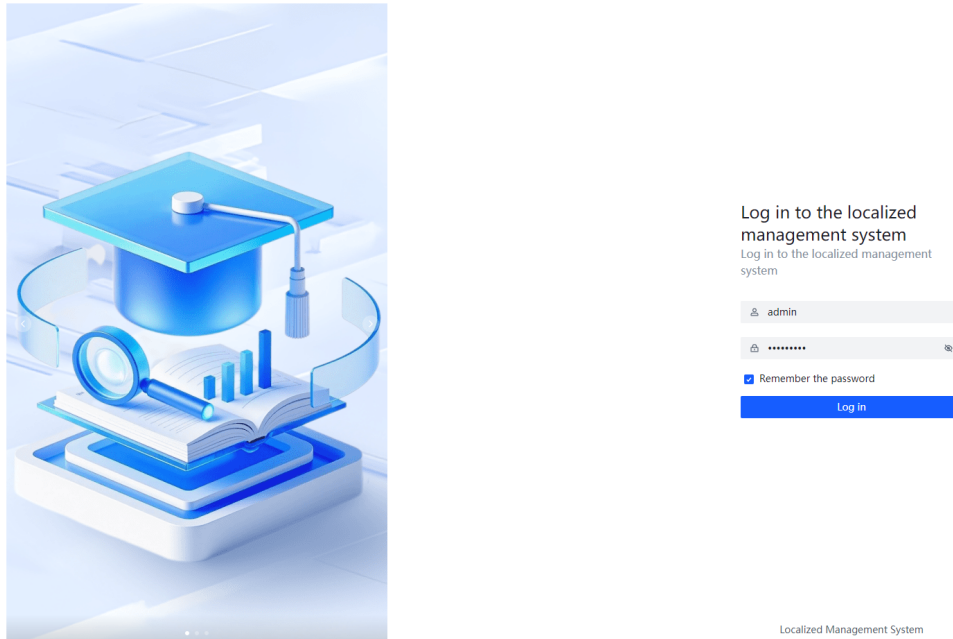


Figure 5: Local Management System - Login

Warning

1. Since devices share the same default IP address from the factory(192.168.1.36), ensure only one device to be configured is connected to the network before changing the IP to avoid IP conflicts.
2. If the device IP has been changed, please log in to the Local Management System using the new IP address.
3. The Local Management System requires a few minutes to initialize after the device is powered on. Please wait for 2 minutes before accessing again.

3.3.2. System Operations and Maintenance

After logging in, click “**System Maintenance**” in the left menu:

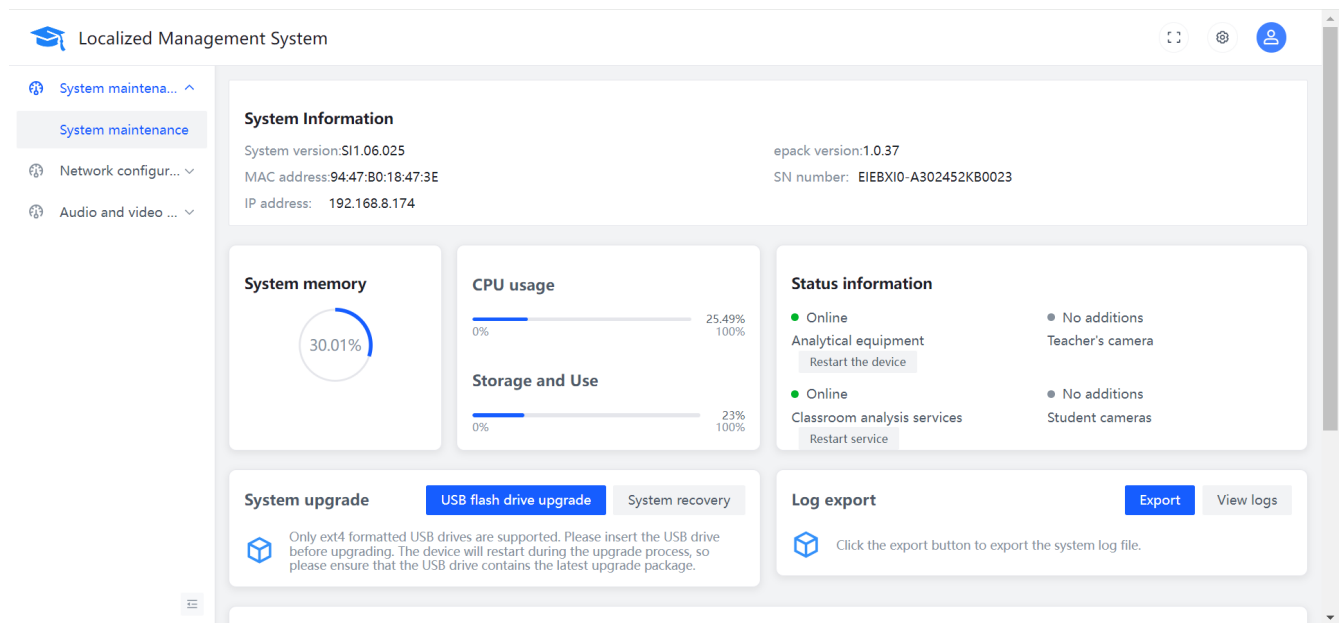


Figure 6: Local Management System - System O&M

1. **System Information:** View basic device information such as the current system version, IP address, SN, and MAC address.
2. **System Memory:** View real-time CPU, memory, and storage usage to assess device health.
3. **Log Management:**
 - **View Logs:** Displays system runtime logs in real-time.
 - **Export Logs:** Packages and downloads logs locally for troubleshooting.
4. **System Time Calibration:**
 - **NTP Calibration:** Enter a valid NTP server address (requires the device to be connected to the public network or an internal NTP service) and click OK to synchronize time.
 - **Manual Calibration:** Click **“Now”** or manually select the date and time, then click OK to save.
5. **Device Upgrade:**
 - **USB Upgrade:** Refer to [Local Management System USB Upgrade](#).
 - **System Restore:** Click **“Restore”** to reset the device to factory settings and clear data.

3.3.3. Network Configuration

Click **“Network Configuration”** in the left menu:

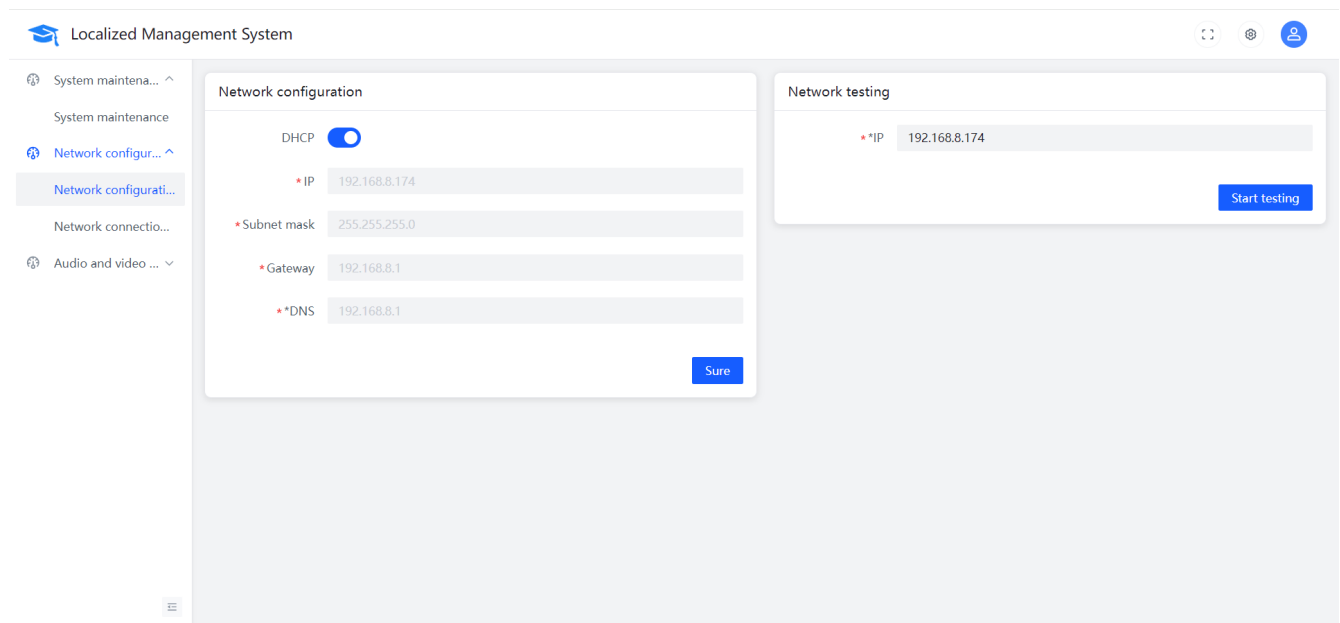


Figure 7: Local Management System - Network Configuration

1. **Modify Static IP:** Uncheck the DHCP option, manually enter the target IP, subnet mask, and gateway. Click OK and re-login using the new IP.
2. **DHCP Automatic Acquisition:** Check the DHCP option, and the device will automatically obtain an available IP in the new network.
3. **Network Test:** Enter a target IP or domain name (e.g., 192.168.1.1) and click Test to verify network connectivity.

4. System Upgrade and Firmware Flashing

This chapter details the system upgrade methods for the EIEBXIO-A30 (IWB), covering Fastboot upgrades, U-Boot USB flash burning, and Web upgrades via the Local Management System.

4.1. Image Download

Please download the corresponding image from the ESWIN official website [Resource Center](#) based on your chosen upgrade method:

- **Path:** Developer Community -> Development Support -> Resource Center -> EIEBXIO-A30 (IWB)
- **File Selection:**
 - For **Fastboot** or **U-Boot** upgrades: Please download `images-SI*.tar.gz`.
 - For **Local Management System** upgrades: Please download `update-SI*.tar.gz`.

4.2. Fastboot Upgrade

This upgrade method is suitable for scenarios where the Bootloader can start normally and system partitions need to be updated quickly.

4.2.1. Preparations

1. **Install Tools:** Windows systems require the installation of fastboot drivers and tools; Linux systems require the installation of fastboot tools.
2. **Download Image:** Download `images-SI*.tar.gz` and extract it to a local directory.
3. **Connect Device:** Use a USB Type-A to Type-A cable to connect Port USB1 to the PC, and use a Type-C data cable to connect the UART serial port.

4.2.2. Windows Driver Installation

4.2.2.1. Tool Download

- [Click to download](#) the Fastboot command-line tool. Extract it and add the `platform-tools` directory to the system's `PATH` environment variable.
- [Click to download](#) the `Android USB Driver for Windows` package for later use.

4.2.2.2. Enter Fastboot Mode and Install Driver

1. Power on the IWB. Press the Enter key during U-Boot startup to enter the U-Boot command line.
2. In the IWB U-Boot command line, enter `fastboot usb0` :

```
=>
=> fastboot usb 0
generic_phy_get_bulk : no phys property
```

A prompt similar to `generic_phy_get_bulk: no phys property` will return on the serial terminal, indicating it is waiting for a connection.

3. An unknown device named `USB download gadget` will appear under “Other devices” in the PC’s “Device Manager” .

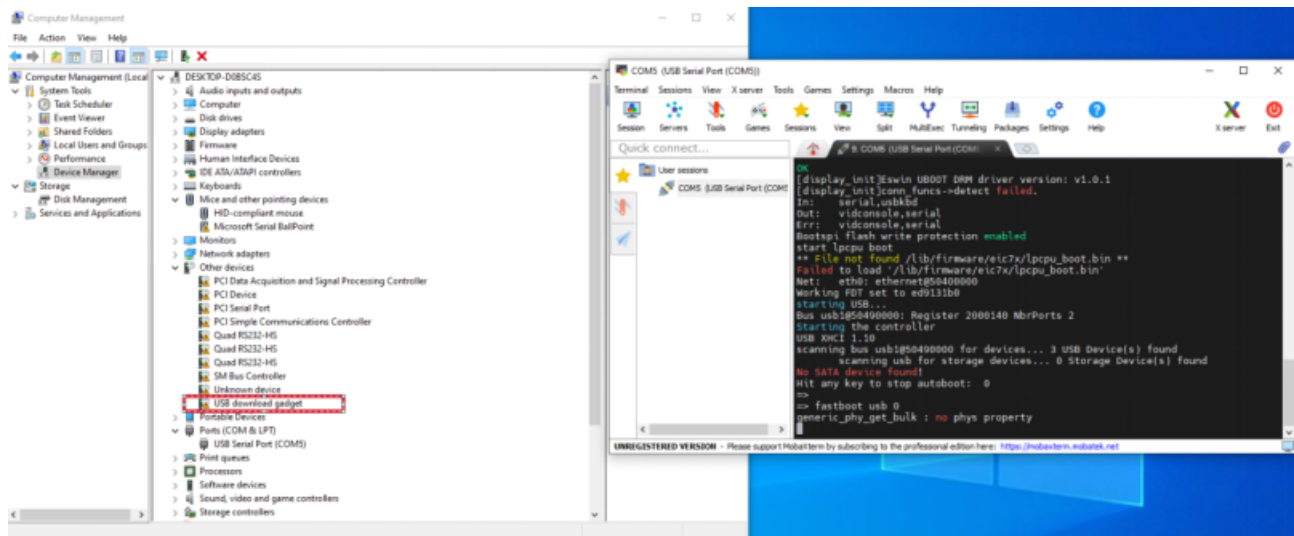


Figure 8: Fastboot Device Port

4. Right-click on the **USB download gadget** device and select **Update driver**.

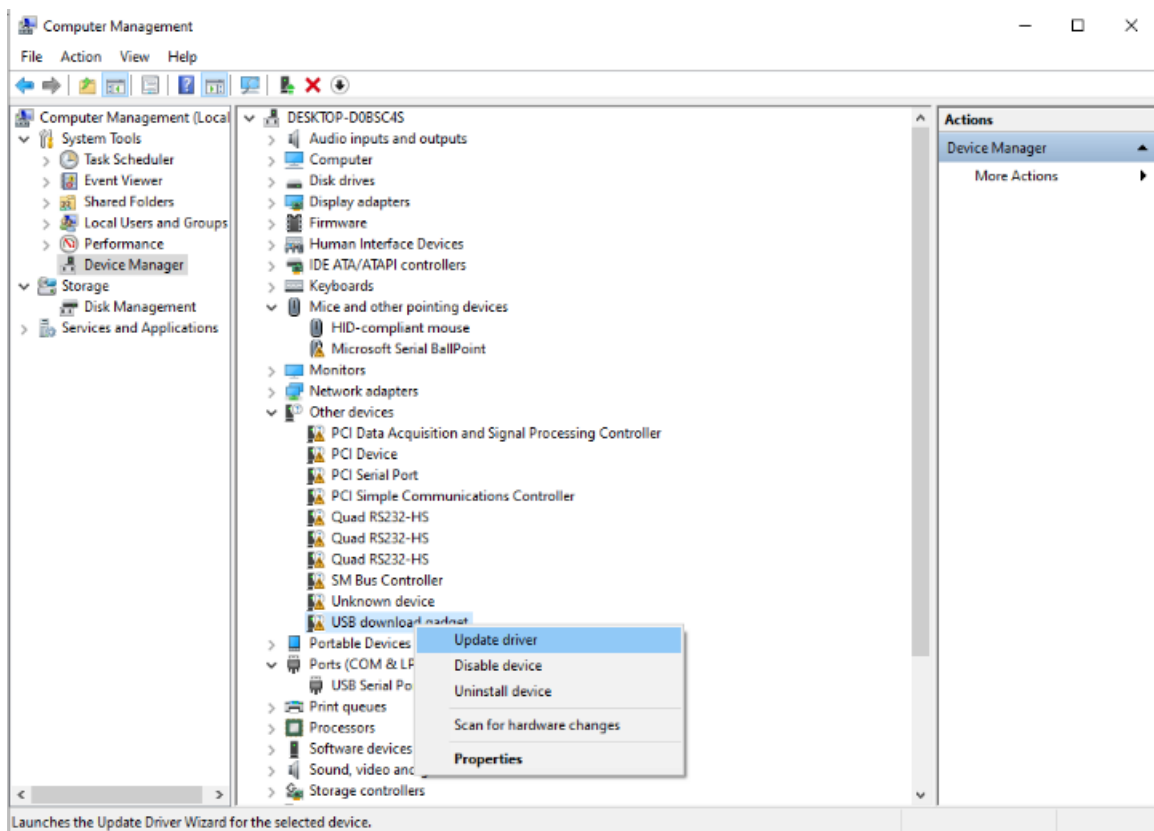


Figure 9: Fastboot Driver Update

5. In the pop-up window, select **Browse my computer for driver software**.

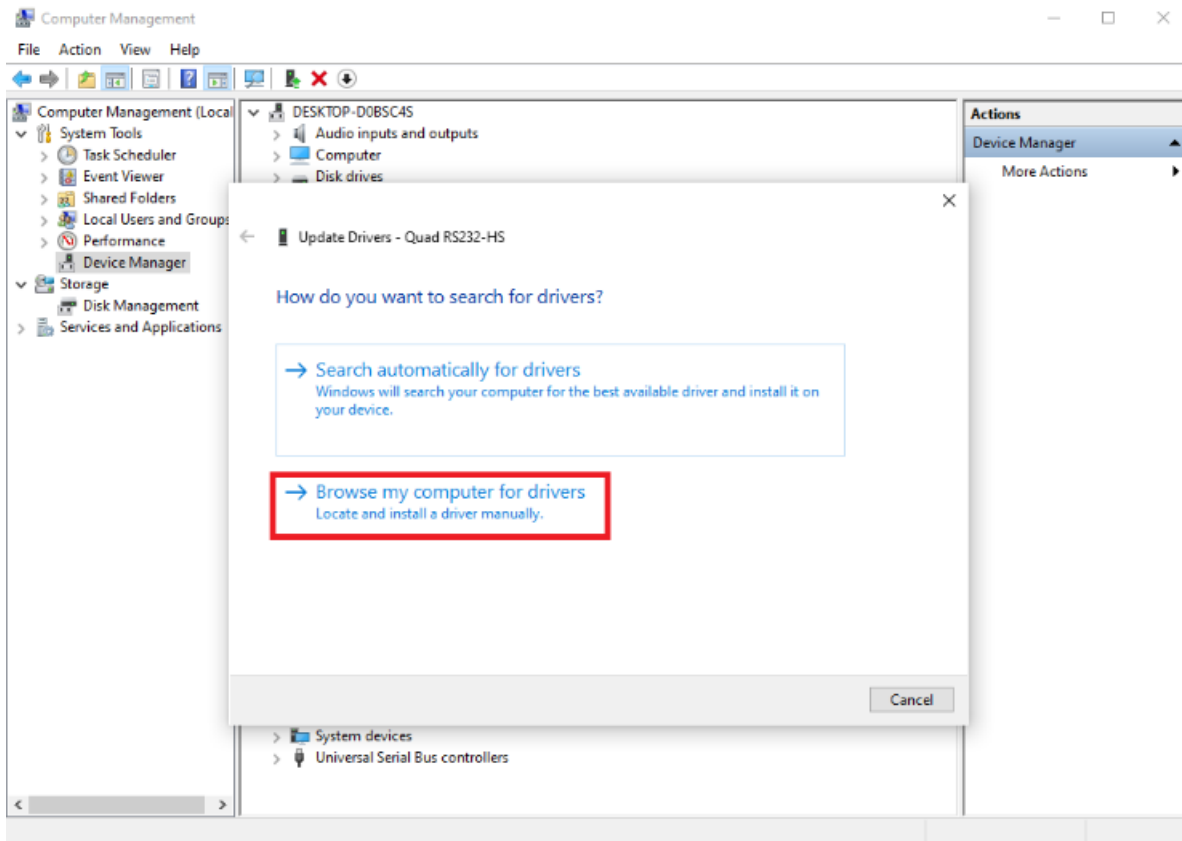


Figure 10: Fastboot Driver Browse

6. Select **Let me pick from a list of available drivers on my computer**.

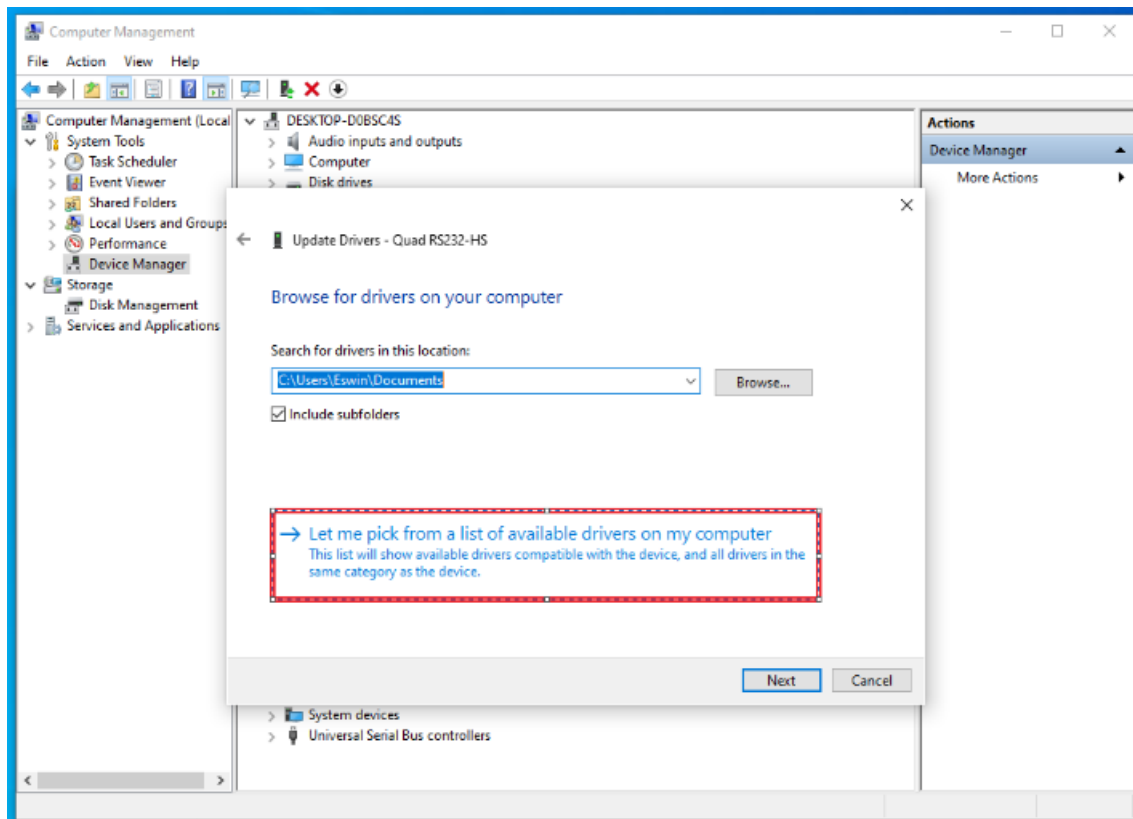


Figure 11: Fastboot Driver Select File

7. In the device type list, select **ADB Interface** and click Next.

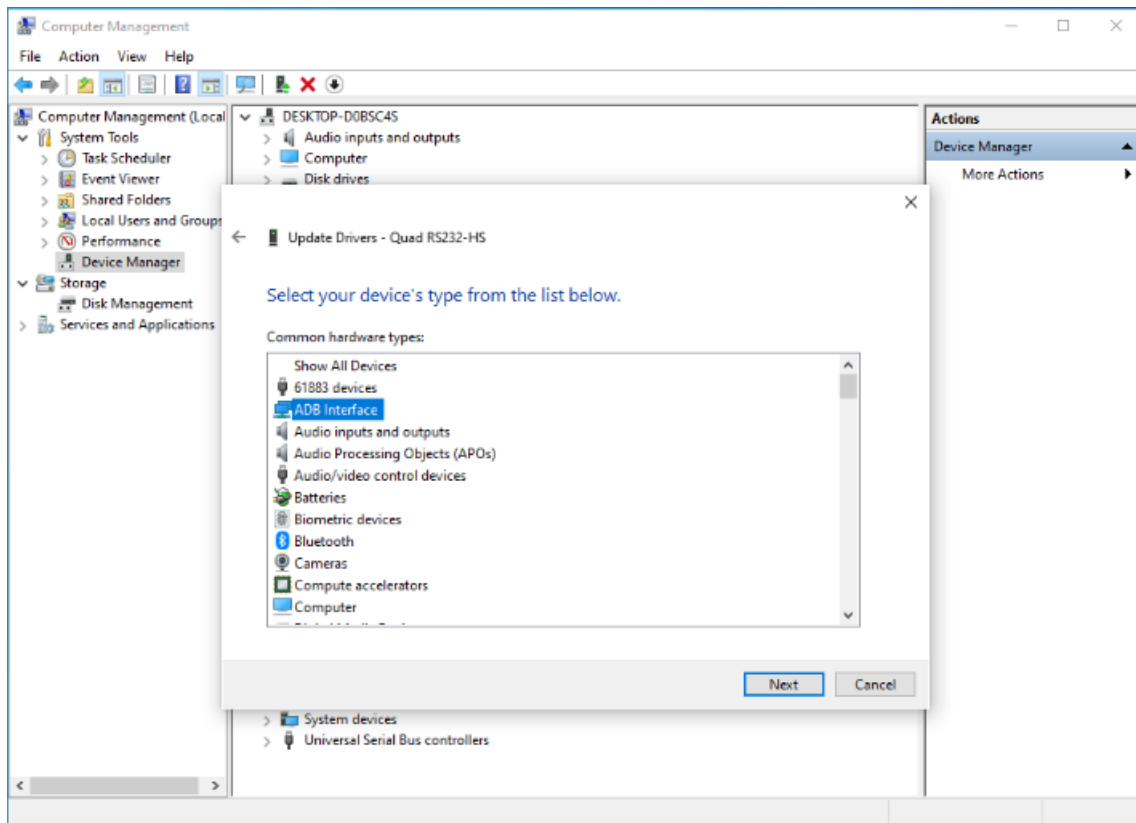


Figure 12: Fastboot Driver Select Device Type

8. Select the first driver and click Next to complete the installation.

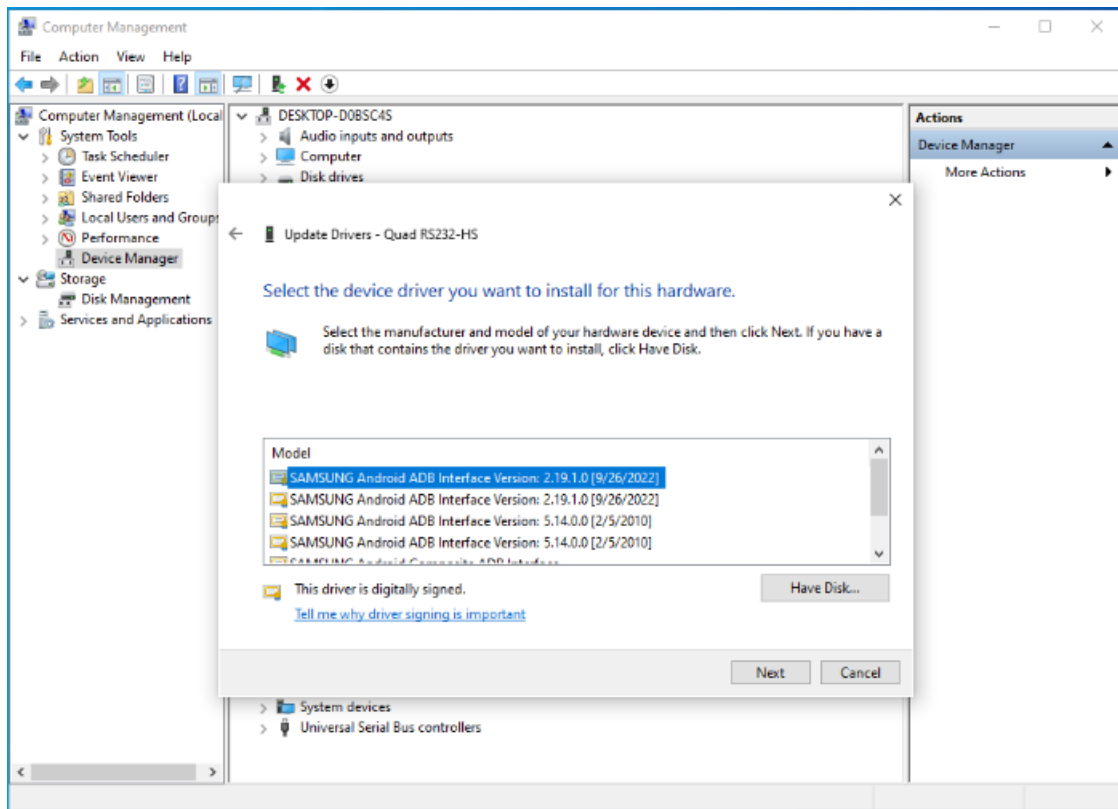


Figure 13: Fastboot Driver Select Driver

9. If the system pops up a warning Windows cannot verify the publisher of this driver software , please select Yes .

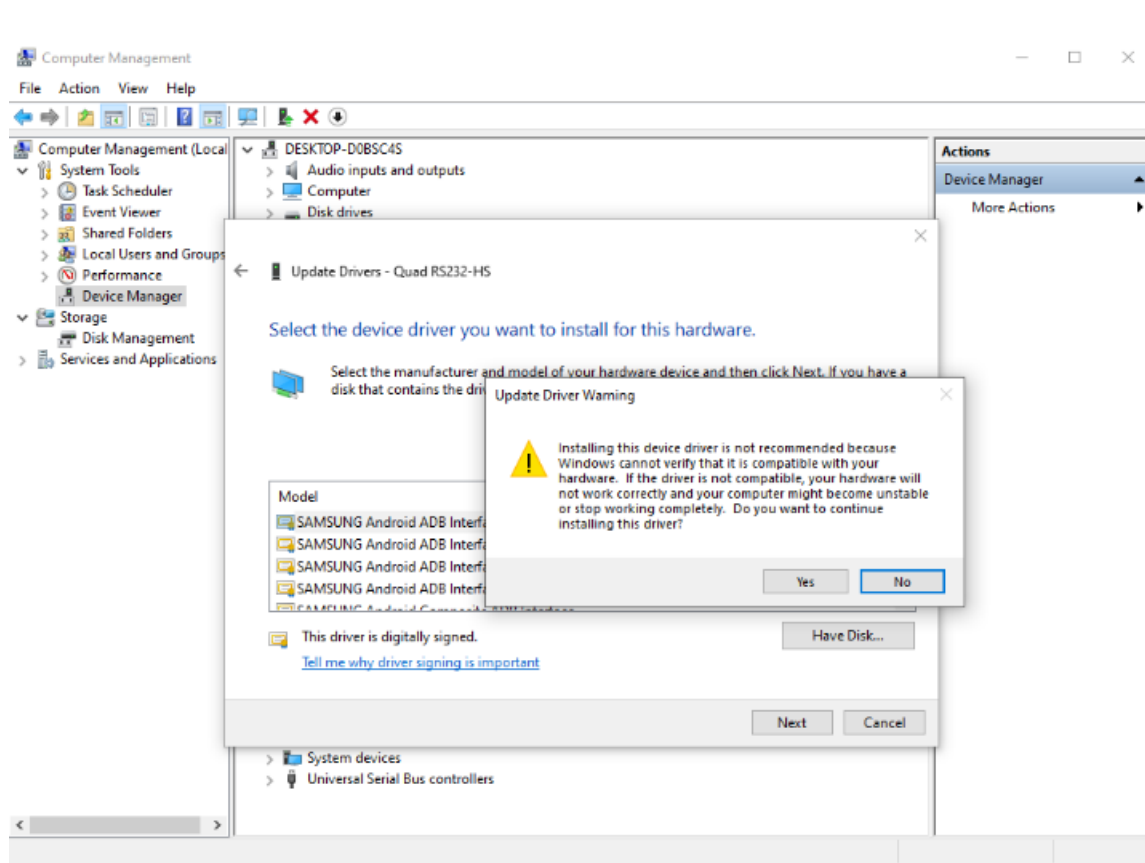


Figure 14: Fastboot Driver Confirmation

4.2.2.3. Verify Fastboot Connection

After the driver installation is successful, a new device should appear under **ADB Interface** in the **Device Manager**.

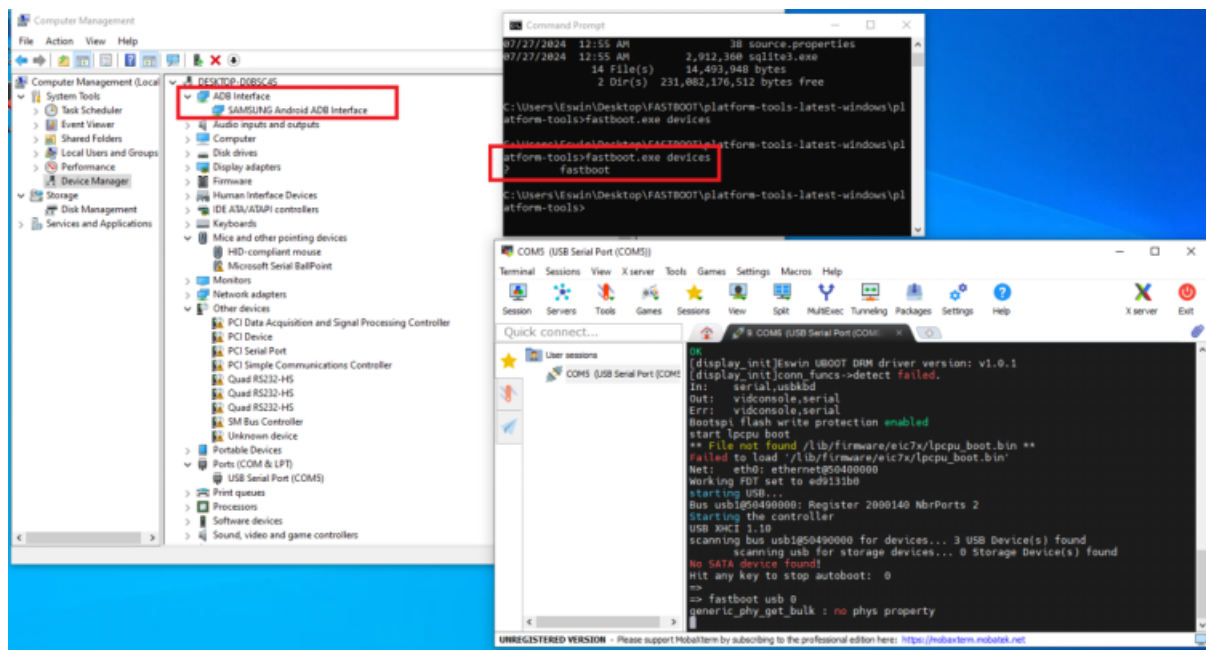


Figure 15: Fastboot Driver Refresh

1. Open the Command Prompt (CMD) or PowerShell on the PC.
2. Enter the following command to check if the device is recognized:

```
fastboot devices
```

3. If the connection is successful, the command line will return the device serial number with the status `fastboot`, for example:

```
xxxxxxx fastboot
```

At this point, the Fastboot driver installation is complete, and you can start using Fastboot commands to flash the system image.

4.2.3. Flash Images

Run the following commands sequentially on the PC (ensure the command line path includes the image files):

```
fastboot flash bootchain bootloader-IWB.bin # Upgrade bootloader
fastboot flash boot boot-IWB.ext4 # Upgrade boot partition
fastboot flash root root-IWB.ext4 # Upgrade root partition
fastboot flash recovery recovery-IWB.ext4 # Upgrade recovery partition
fastboot reboot # Reboot device
```

Tip

1. Before upgrading the system image, please ensure:
 - The Fastboot driver is installed.
 - The device is in Fastboot mode.
 - The connection between the PC and IWB is correct.
2. If you need to upgrade a single partition, simply execute the corresponding fastboot command based on the partition name.

3. When upgrading the Boot and Root partitions, the corresponding partitions will be re-flashed and overwritten. Please back up the data you need to the Data partition or external storage device in advance to avoid data loss.

4.3. U-Boot USB Upgrade

This method requires copying the image files to a USB drive first, and then completing the system upgrade via U-Boot serial commands. The commands are simple and easy to use.

4.3.1. Preparation

1. **Prepare the medium:** Prepare a USB flash drive with a capacity of at least 16GB and format it as a single ext4 partition.
2. **Copy images:** Download and extract `images-SI*.tar.gz`, then copy the extracted image files directly to the root directory of the USB flash drive. (Note: Do not place the files in any subfolders.)
3. **Connect the device:** Since the IWB only supports peripherals on the USB 2.0 port in Bootloader mode, be sure to insert the USB flash drive into the USB 2.0 port.
4. **Connect to serial:** Use a Type-C data cable to connect to the device's UART serial port.

4.3.2. Upgrade Procedure

1. Power on the IWB. Press the **Enter** key during U-Boot startup to enter the U-Boot command line.
2. Check the loading status: The message `1 Storage Device(s) found` indicates a successful load. If the USB flash drive is not successfully recognized, execute the `usb reset` command to rescan the device.

```
=> usb reset
resetting USB...
Bus usb1@50490000: Register 2000140 NbrPorts 2
Starting the controller
USB XHCI 1.10
scanning bus usb1@50490000 for devices... 2 USB Device(s) found
scanning usb for storage devices... 1 Storage Device(s) found
```

3. Verify USB files: List the directory contents and ensure the upgrade files are present in the USB root directory (boot-IWB.ext4、bootloader-IWB.bin、recovery-IWB.ext4、root-IWB.ext4 以及 usbupdate.scr).

```
=> ls usb 0
<DIR> 4096 .
<DIR> 4096 ..
524288000 boot-IWB.ext4
4393384 bootloader-IWB.bin
5368709120 recovery-IWB.ext4
7516192768 root-IWB.ext4
201 SI1.06.025-md5
1455 update-usb.txt
1527 usbupdate.scr
```

4. Execute the upgrade: Wait for the prompt `The burn script execute complete` after running the upgrade command, which indicates completion.

```
=> run usbupdate
1527 bytes read in 6 ms (248 KiB/s)
```

```
## Executing script at 90000000
## Resetting to default environment
[VO][INFO] display_init:480 Die[0] Eswin UBOOT DRM driver version: v1.0.1
[VO][INFO] display_init:518 no display connected!
Saving Environment to SPIFlash... Erasing SPI flash...Writing to SPI flash...done
OK
save env successfully
Ready to partition
Writing GPT: success!
Set partitions successfully
Ready to burn bootloader
4395352 bytes read in 42 ms (99.8 MiB/s)
Bootspi flash write protection disabled
SF: 4096 bytes @ 0x1000 Read: OK
Erase progress: 100%:+++++
SF: 4096 bytes @ 0x1000 Erased: OK
Write progress: 100%:+++++
SF: 0x1000 bytes @ 0x1000 Written: OK
SF: 4096 bytes @ 0x0 Read: OK
FIRMWARE writing...
Erase progress: 100%:+++++
SF: 2044 bytes @ 0x50000 Erased: OK
Write progress: 100%:+++++
SF: 0x7fc bytes @ 0x50000 Written: OK
DDR writing...
Erase progress: 100%:+++++
SF: 455808 bytes @ 0x51000 Erased: OK
Write progress: 100%:+++++
SF: 0x6f480 bytes @ 0x51000 Written: OK
BOOTLOADER writing...
Erase progress: 100%:+++++
SF: 3936696 bytes @ 0x10e000 Erased: OK
Write progress: 100%:+++++
SF: 0x3c11b8 bytes @ 0x10e000 Written: OK
BOOTCHAIN HEAD writing...
Erase progress: 100%:+++++
SF: 4096 bytes @ 0x0 Erased: OK
Write progress: 100%:+++++
SF: 0x1000 bytes @ 0x0 Written: OK
bootloader write OK
Bootspi flash write protection enabled
Burn bootloader successfully
Ready to burn boot partition
Write progress: 100%:+++++
mmc has been successfully written in mmc 0:1
Burn boot successfully
Ready to burn recovery partition
Write progress: 100%:+++++
mmc has been successfully written in mmc 0:3
Burn recovery successfully
Ready to burn root partition
Write progress: 100%:+++++
mmc has been successfully written in mmc 0:4
Burn root successfully
The burn script execute completely, please power off and switch to spiboot
mode[SW:0010], then power on.
```

Tip

During the system upgrade, the Boot and Root partitions will be reflashed and overwritten. If you have files that you do not want to lose, please save them to the Data partition or an external hard drive in advance to avoid data loss.

5. After the upgrade is complete, reboot the device.

=> reset

4.4. Local Management System Web Upgrade

In addition to physical burning, users can also perform firmware upgrades via the Web interface. This method is intuitive and suitable for rapid on-site updates.

4.4.1. Preparation

1. **Prepare the Media:** Prepare a USB flash drive with a capacity of at least 16GB, formatted as a single ext4 partition.
2. **Download and Copy Image:** Download the `update-SI*.tar.gz` file, extract it, and copy the files directly to the root directory of the USB drive (do not place them inside any subdirectories).
3. **Insert Device:** Insert the USB drive into any USB port on the IWB device.
4. **Configure Network:** Ensure that the PC and the IWB device are on the same local area network (default subnet: `192.168.1.x`).

4.4.2. Upgrade Procedure

1. **Log in to the System**
 - Enter the default address `http://192.168.1.36` in your browser.
 - Enter Credentials: Input the username `admin` and password `Admin@123`, then click Login.
2. **Navigate to the Upgrade Page**
 - After logging in, you will be directed to the “System Maintenance” page by default.
3. **Initiate Upgrade**
 - At the bottom of the page and locate the “System upgrade” section.
 - Ensure the USB drive is properly inserted into the device.
 - Click the blue “**USB flash driver upgrade**” button.
 - In the “System Upgrade Confirmation” pop-up window, click the “**Start Verification**” button.
 - Once the system displays the file verification passed prompt, click the “**Sure**” button to initiate the upgrade process.

Tip

If you encounter a “USB not recognized” error, please check the following:

- Ensure the USB drive is formatted correctly (ext4 format).
- Confirm that the upgrade files are located in the root directory of the USB drive.
- Verify that the upgrade files on the USB drive are correct (please refer to the image below).

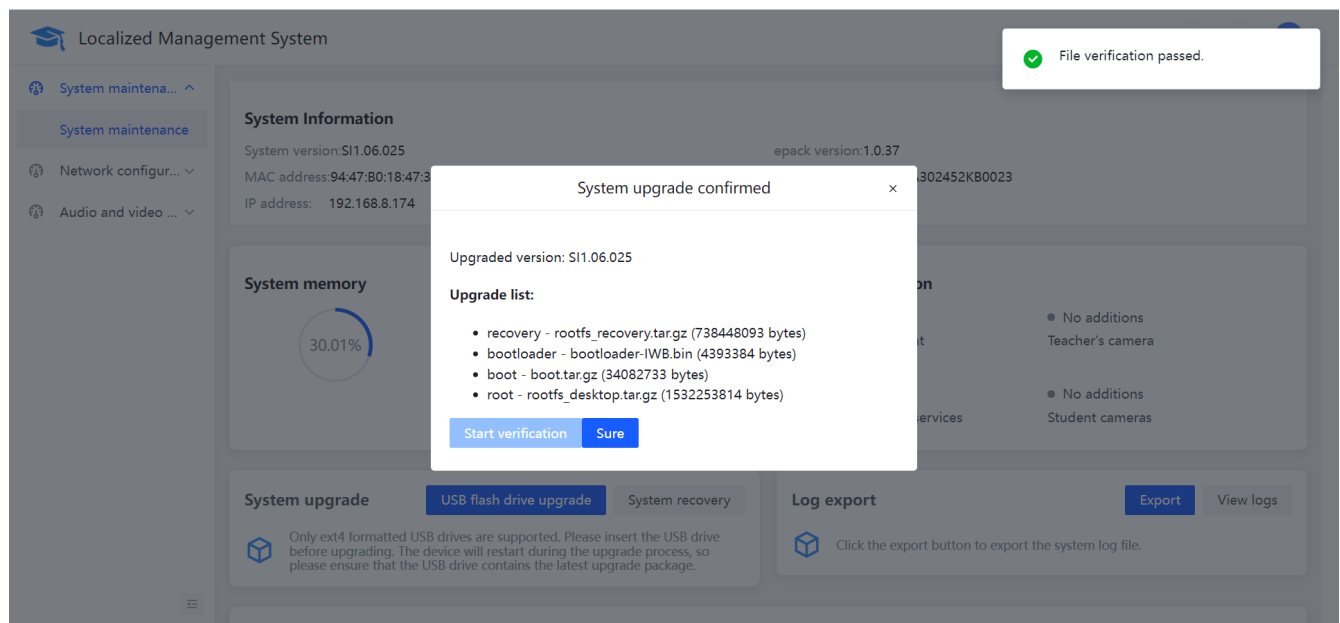


Figure 16: Upgrade

4. Wait and Validate

- **Wait for Reboot:** The device will automatically restart during the upgrade. The Web page will temporarily disconnect. Do not cut the power or unplug the USB drive.
- **Re-login:** Attempt to log in again after approximately 20 minutes.
- **Check Versions:** Check the “System Information” section at the top of the page to confirm that both the **System Version** has been updated to the latest releases.